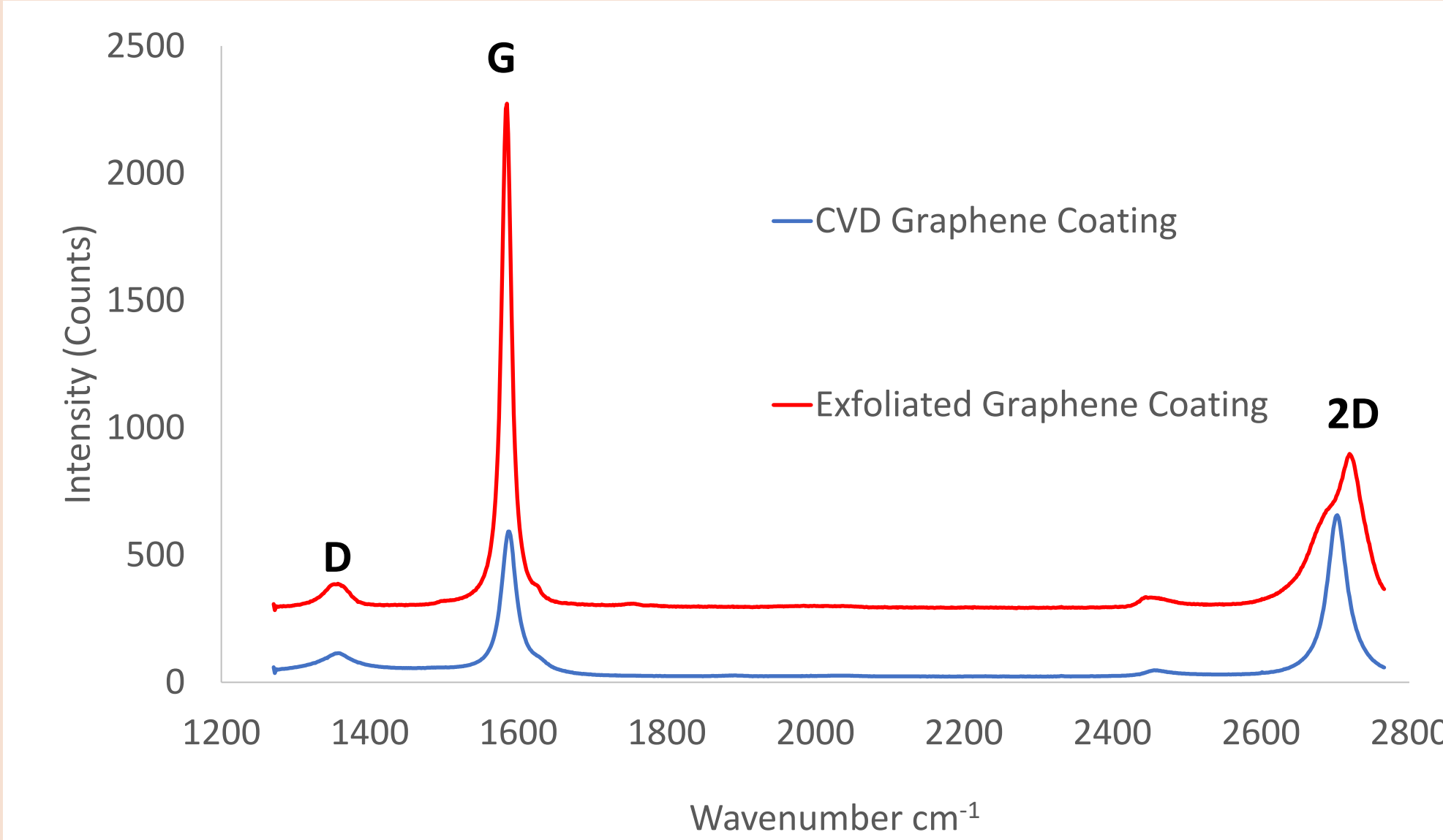


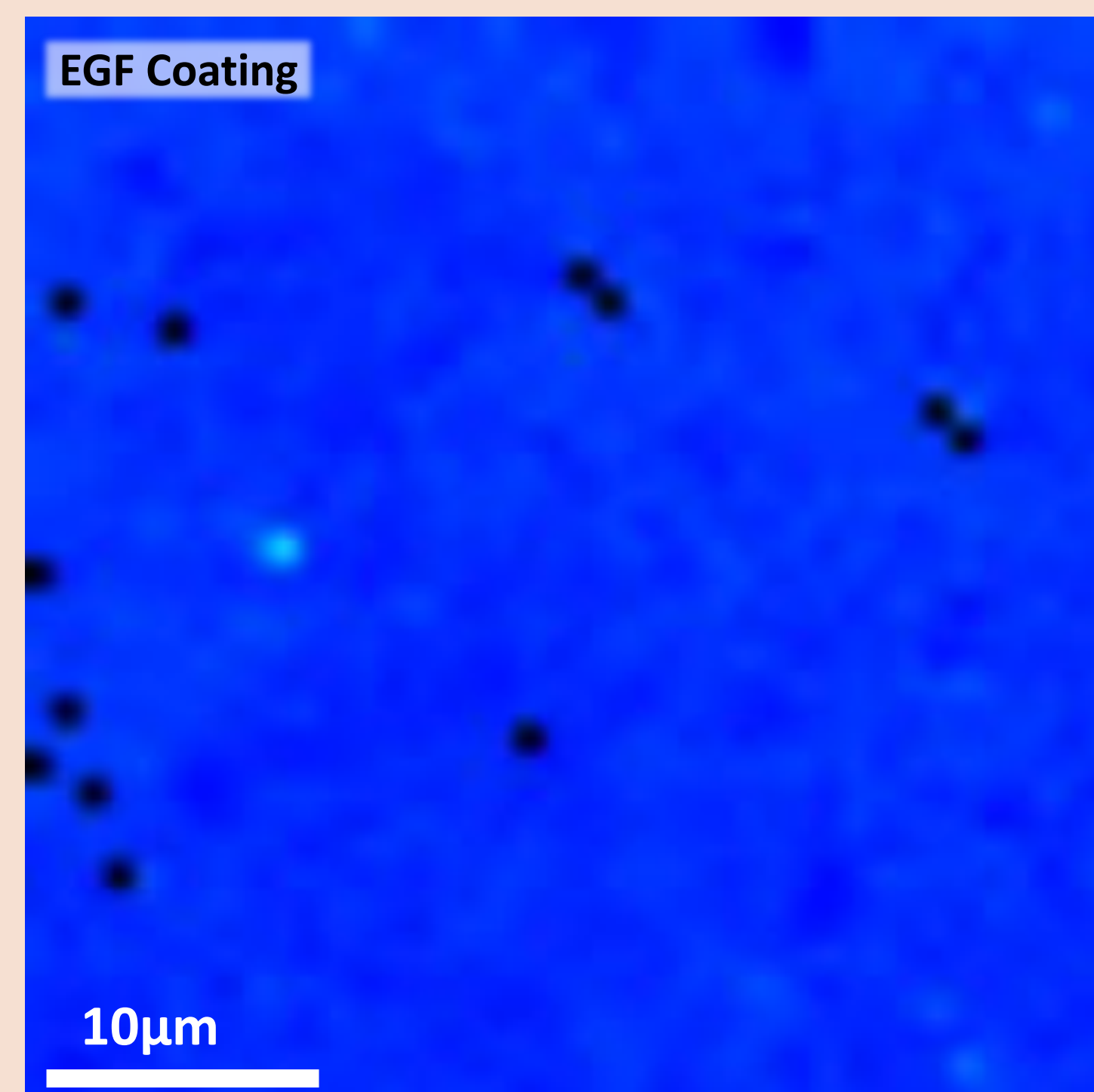
Introduction

- The 8620 alloy is an iron-rich, low-carbon steel, with a martensitic microstructure when produced via additive manufacturing (AM).
- Iron-rich alloys are susceptible to corrosion; therefore, two methods of graphene coating were developed:
 - Exfoliated graphene flakes (EGFs) applied as a thin film coating.
 - Chemical vapor deposition (CVD) graphene grown directly on the metal surface.
- Characterization of the graphene coatings was done via optical microscopy, SEM, and Raman spectroscopy. Corrosion testing methods show promising results.

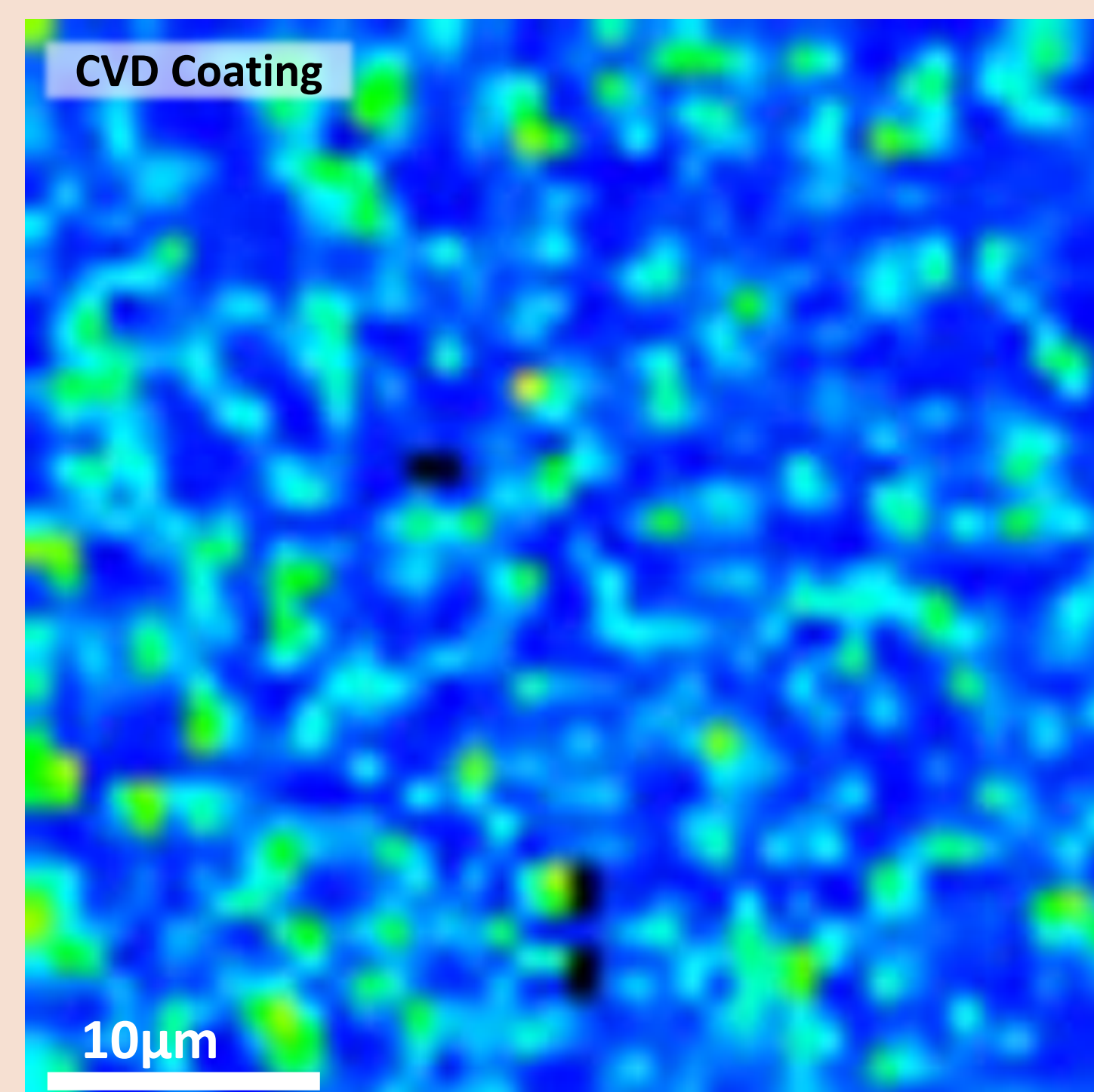
Raman Spectroscopy of Graphene Coatings



- Raman spectroscopy is one of the key methods used to determine the thickness and crystalline qualities of graphene. The 2D:G ratio is greater than 1 for monolayer graphene, and less than 1 for multilayer graphene. The intensity of a D peak corresponds to defects in the graphene crystal lattice.



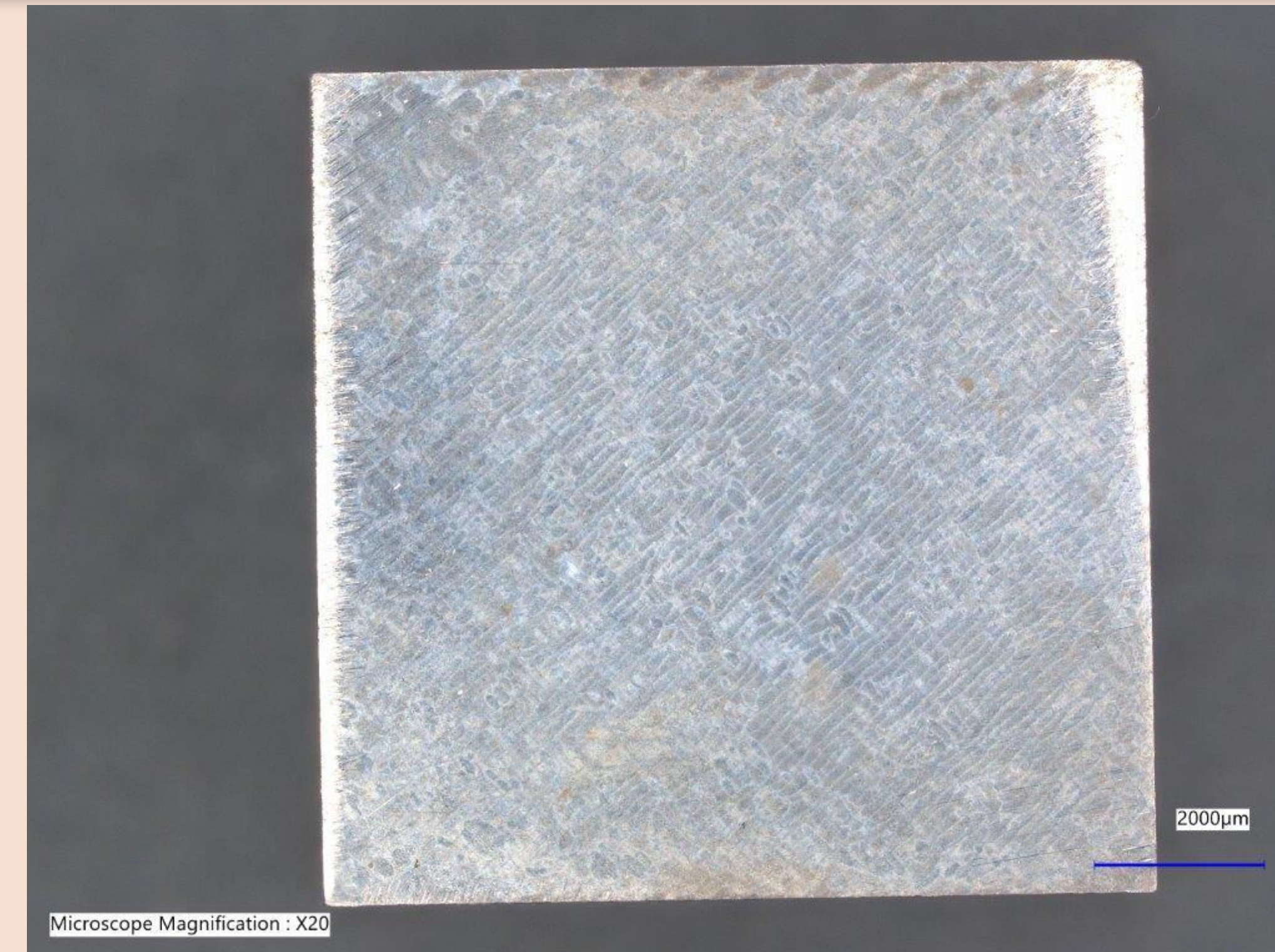
- The average 2D:G ratio for the multilayer EGF coating was 0.26.



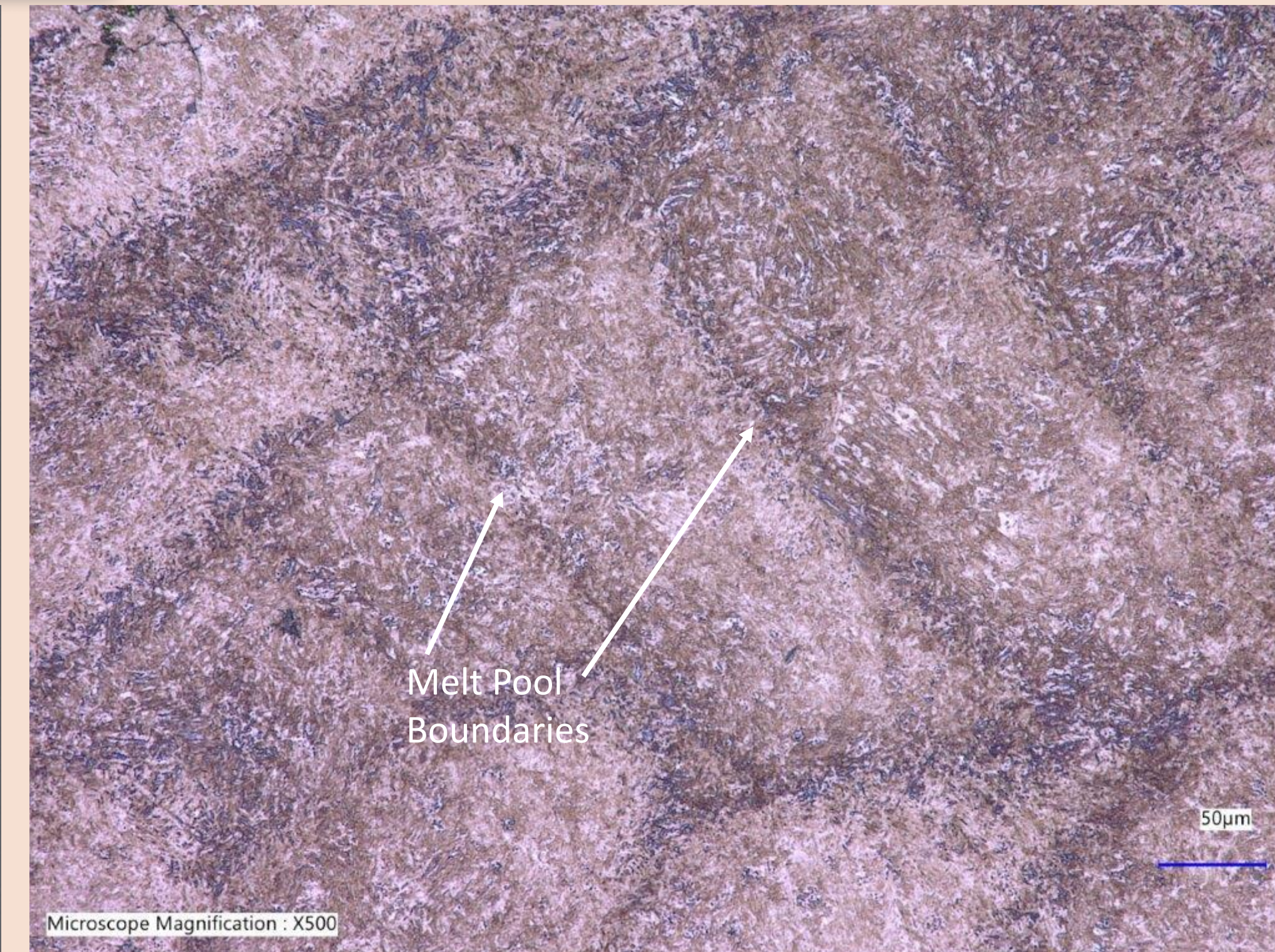
- The average 2D:G ratio for the primarily monolayer CVD graphene coating was 1.2



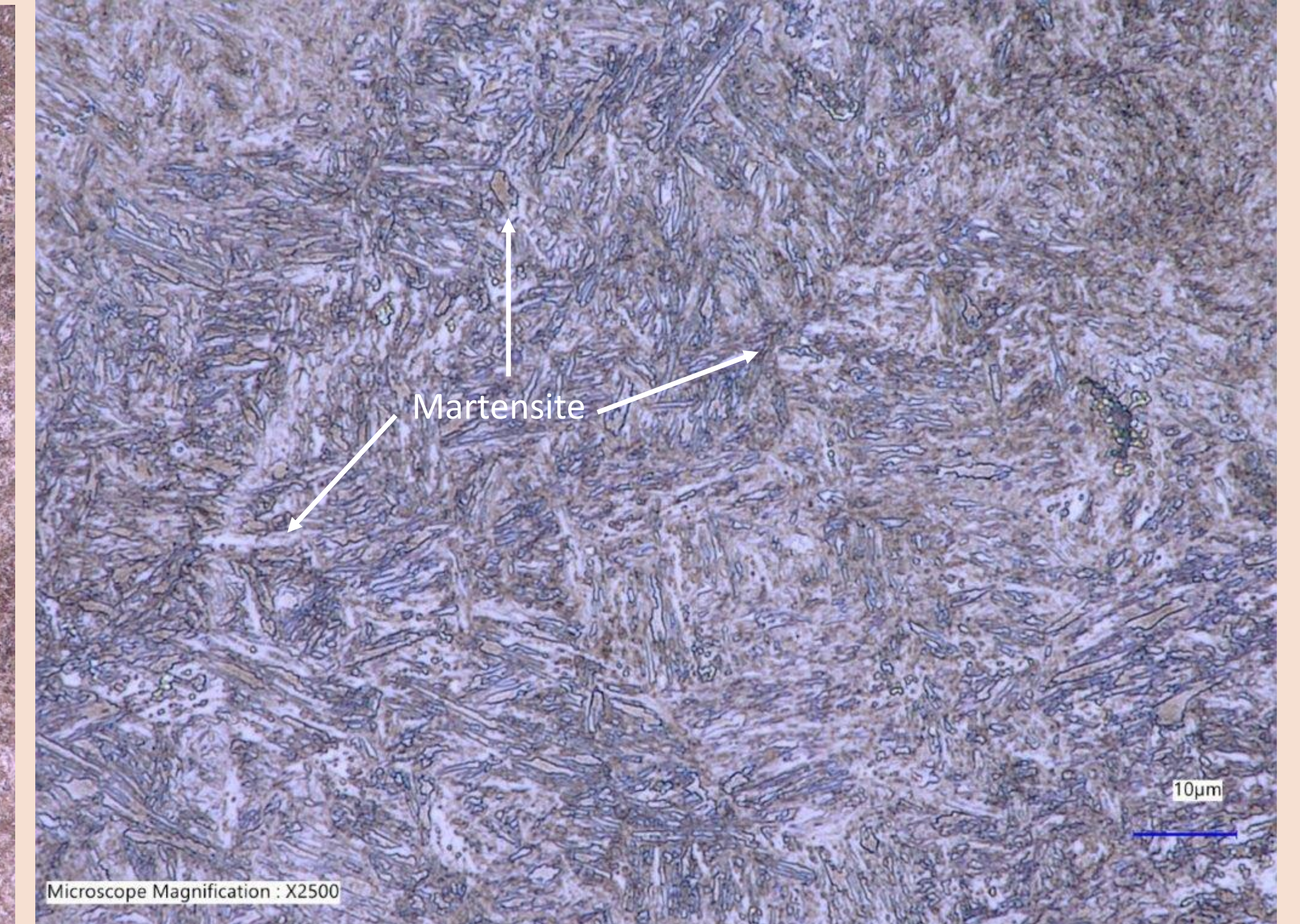
Optical Microscopy of AM 8620 Microstructure



Low magnification image showing the polished and chemically etched surface of an AM8620 cube.. The 90° laser scanning pattern can be seen across the surface of the cube.

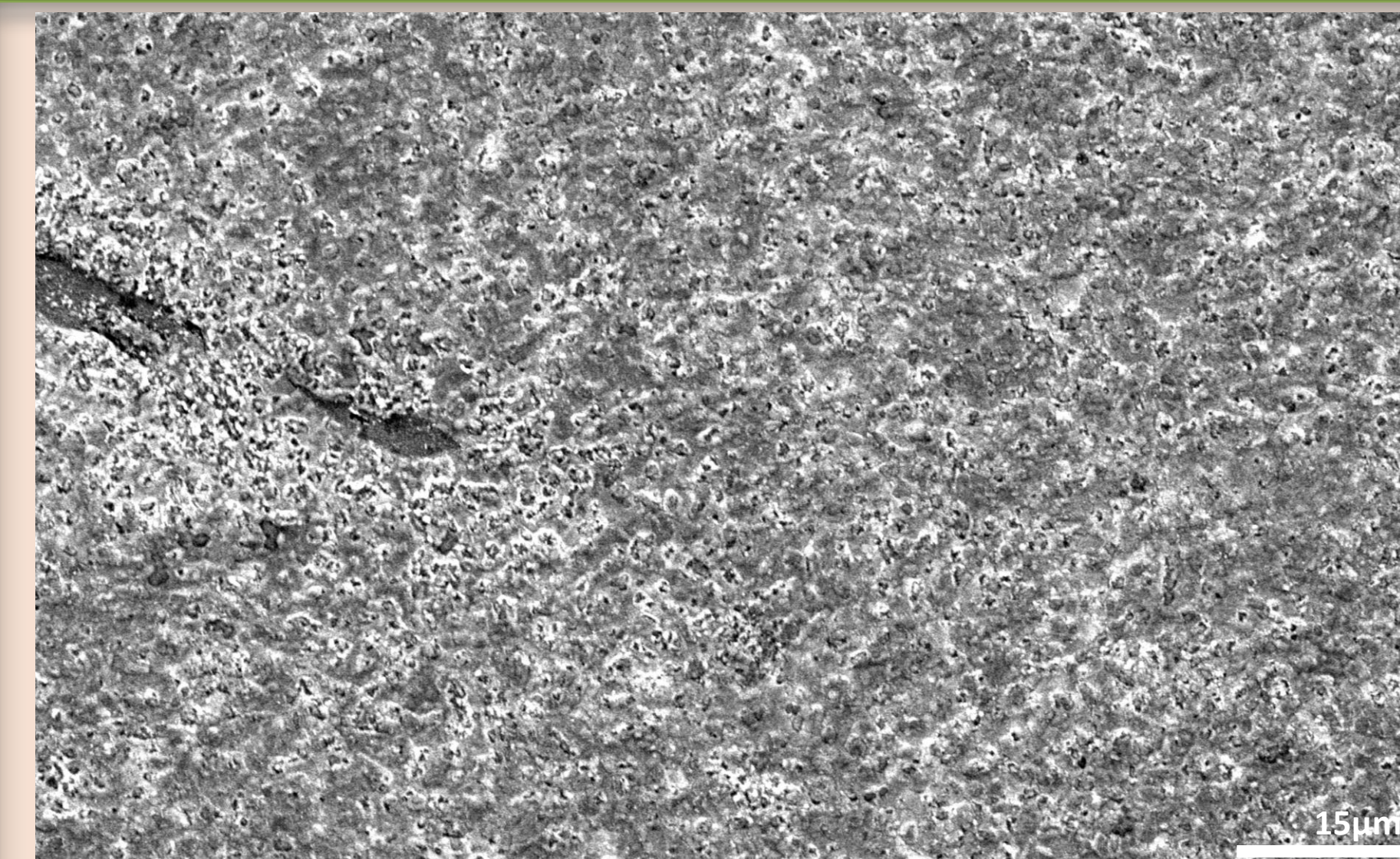


At higher magnification the melt pool boundaries can be clearly seen, and the 90° scanning pattern used for this sample.

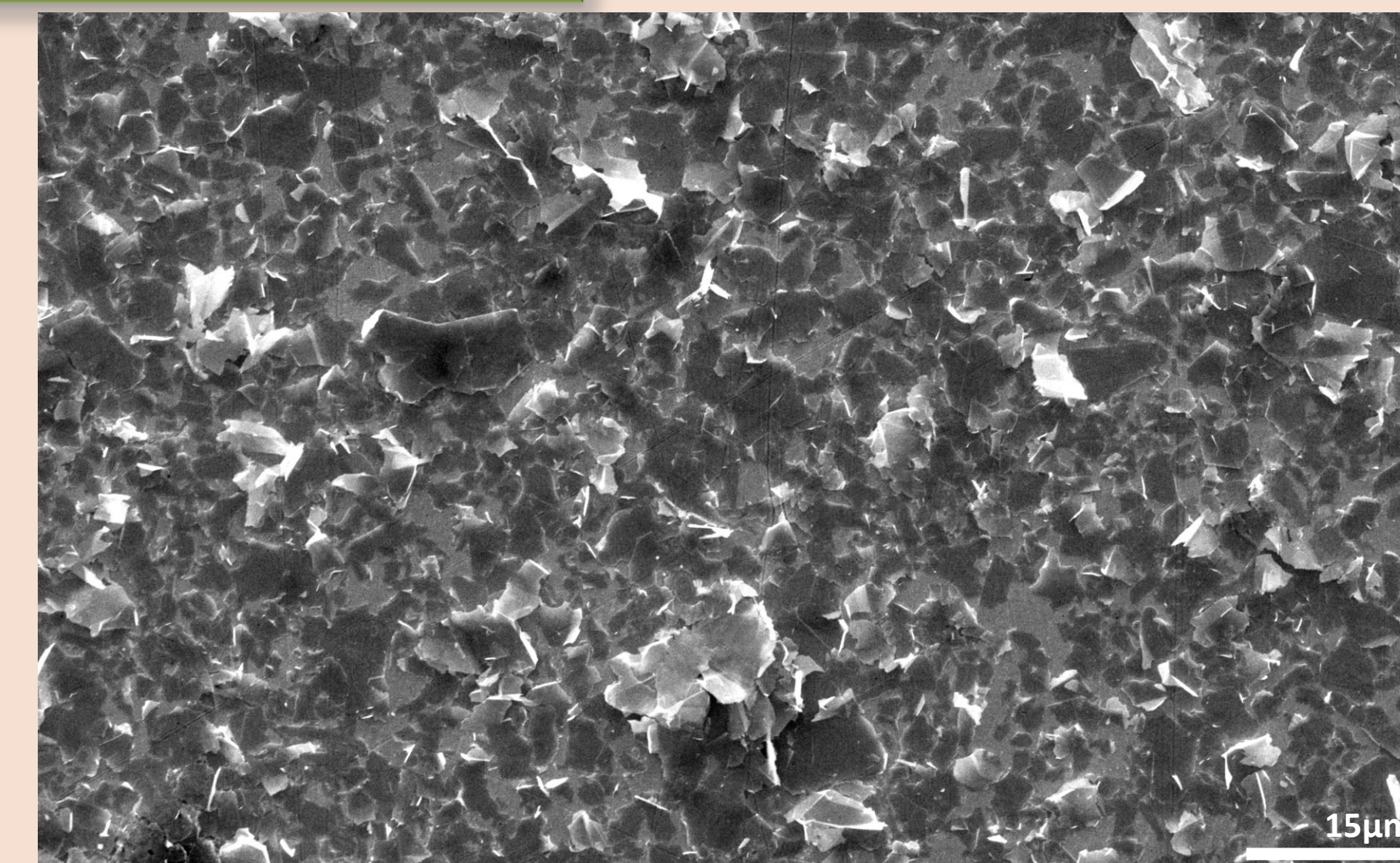


At the highest magnification, the martensitic microstructure of the AM8620 can be seen. This makes the material harder than conventional manufacturing methods, and the uniform structure makes it optimum for applying graphene coatings.

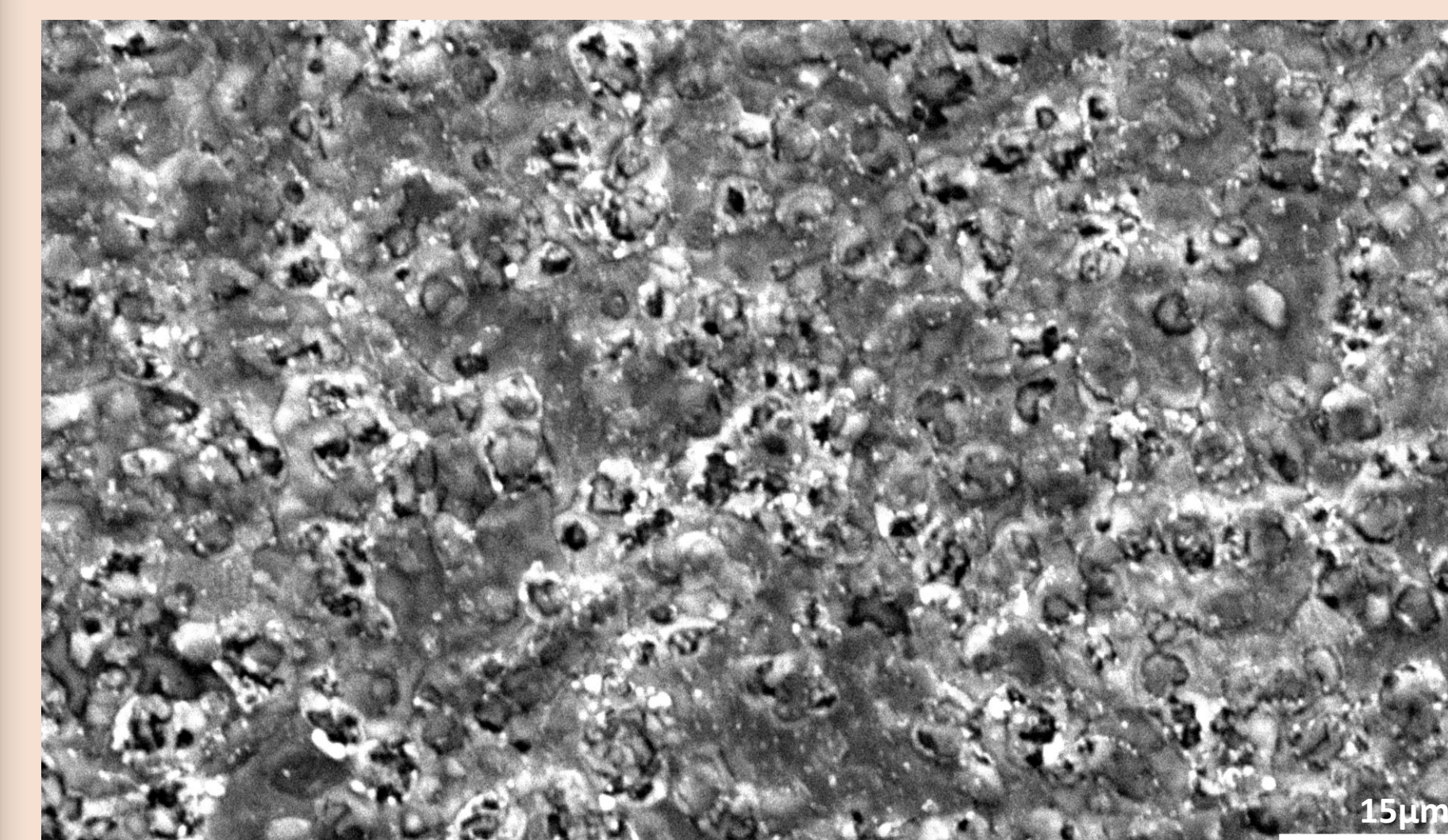
Scanning Electron Microscopy of Graphene Coatings



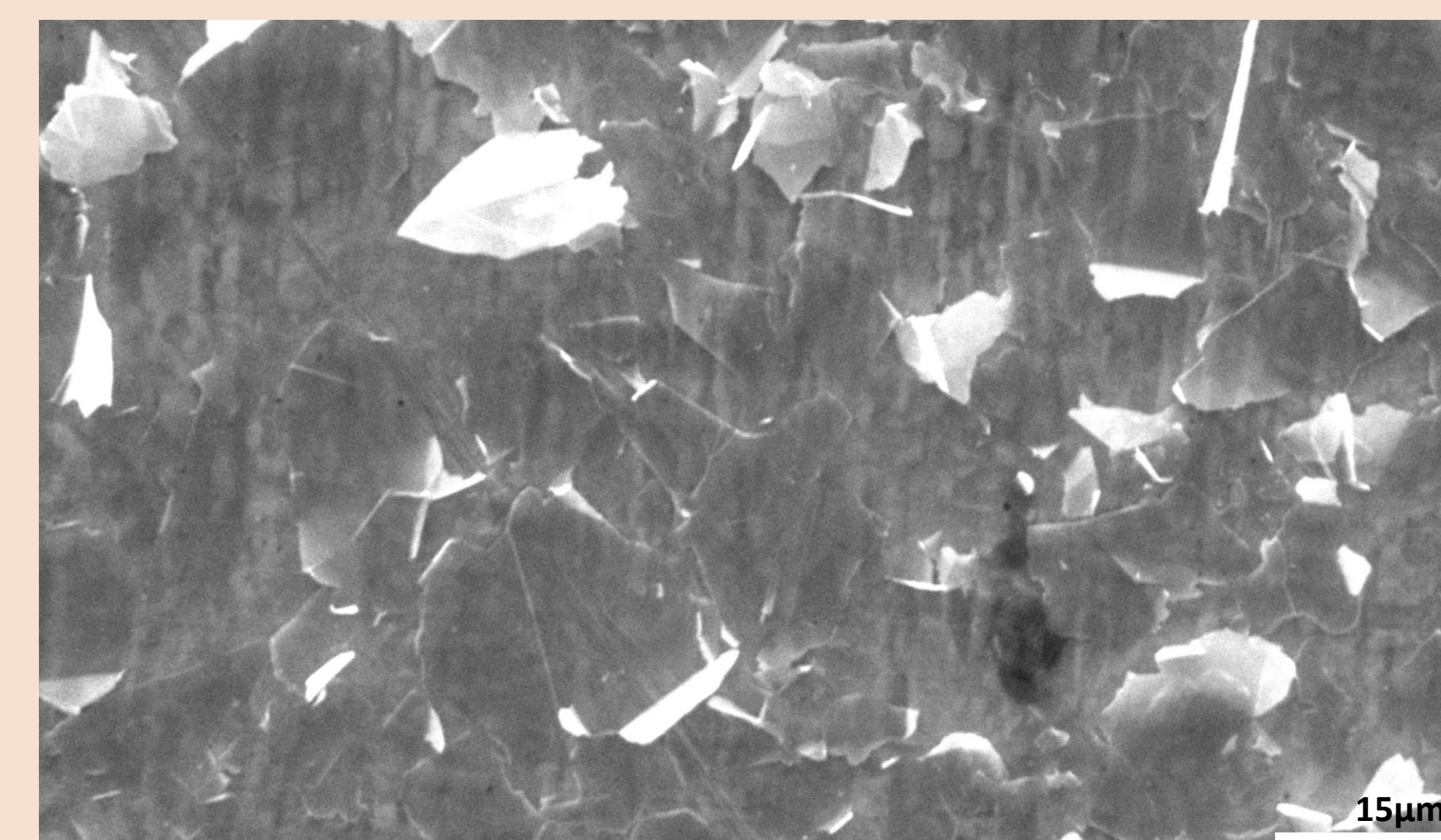
CVD graphene coating on the AM8620 sample with high conformal coverage.



EGF coating on the AM8620 sample with large area coverage.

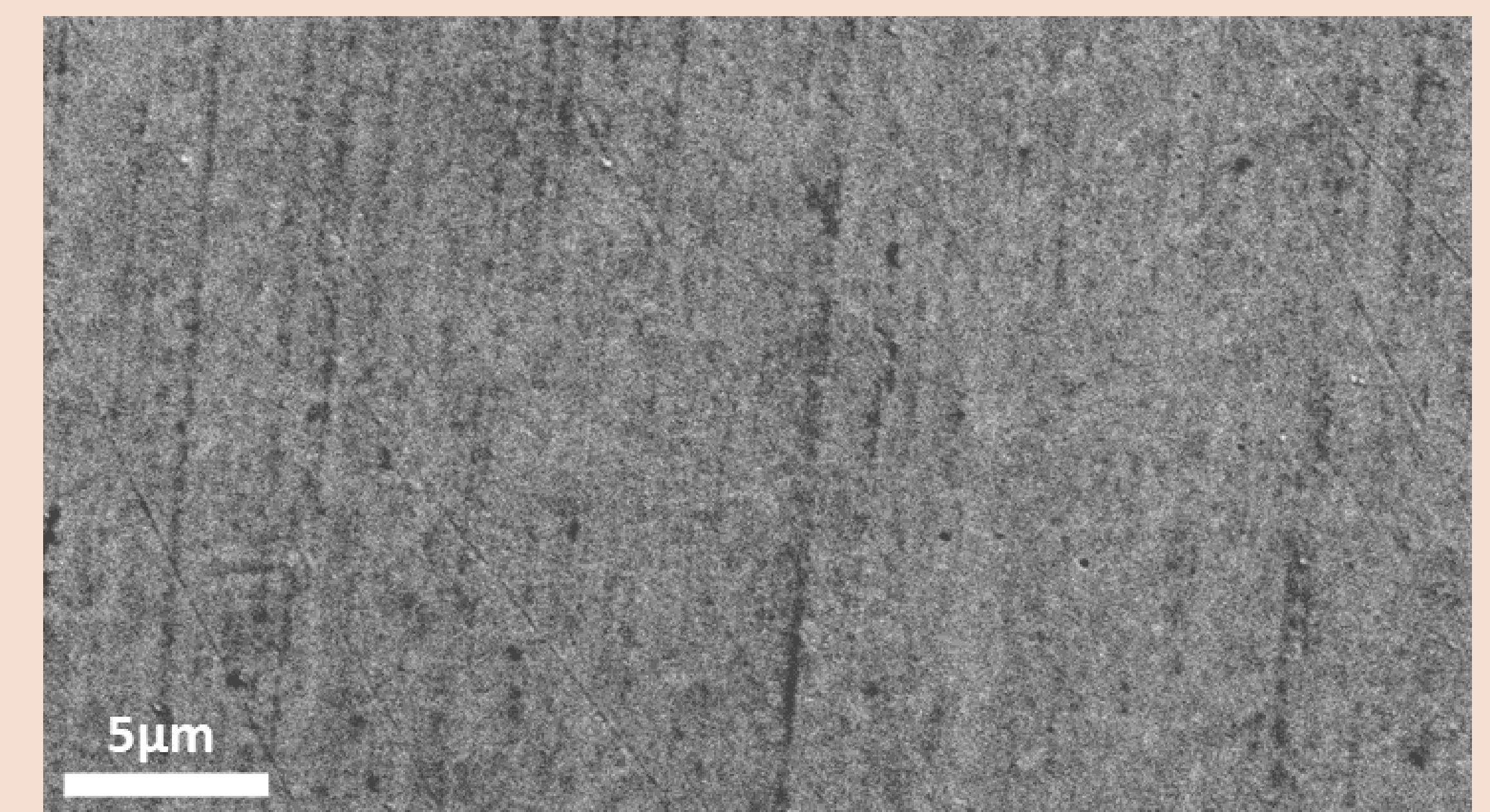


Detail on the CVD coating, showing graphene islands which expand to cover the substrate.



Detail on the EGF coating, showing overlapping flakes.

- Surface preparation by polishing is important to improve adhesion of the EGF coatings and improve uniform nucleation and growth of CVD graphene.
- The EGF coating was capable of coating large areas (~5cm²) with large flakes for high coverage, and small flakes filling the gaps. Note the EGF coating is still thin enough to be transparent to the electron beam.
- The microstructure (martensite) strongly improves the CVD graphene coating, since the grains and composition are more uniform than conventional carbon steel, accommodating better nucleation and uniform growth. The CVD coating could cover some cracks and defects found in the AM material.

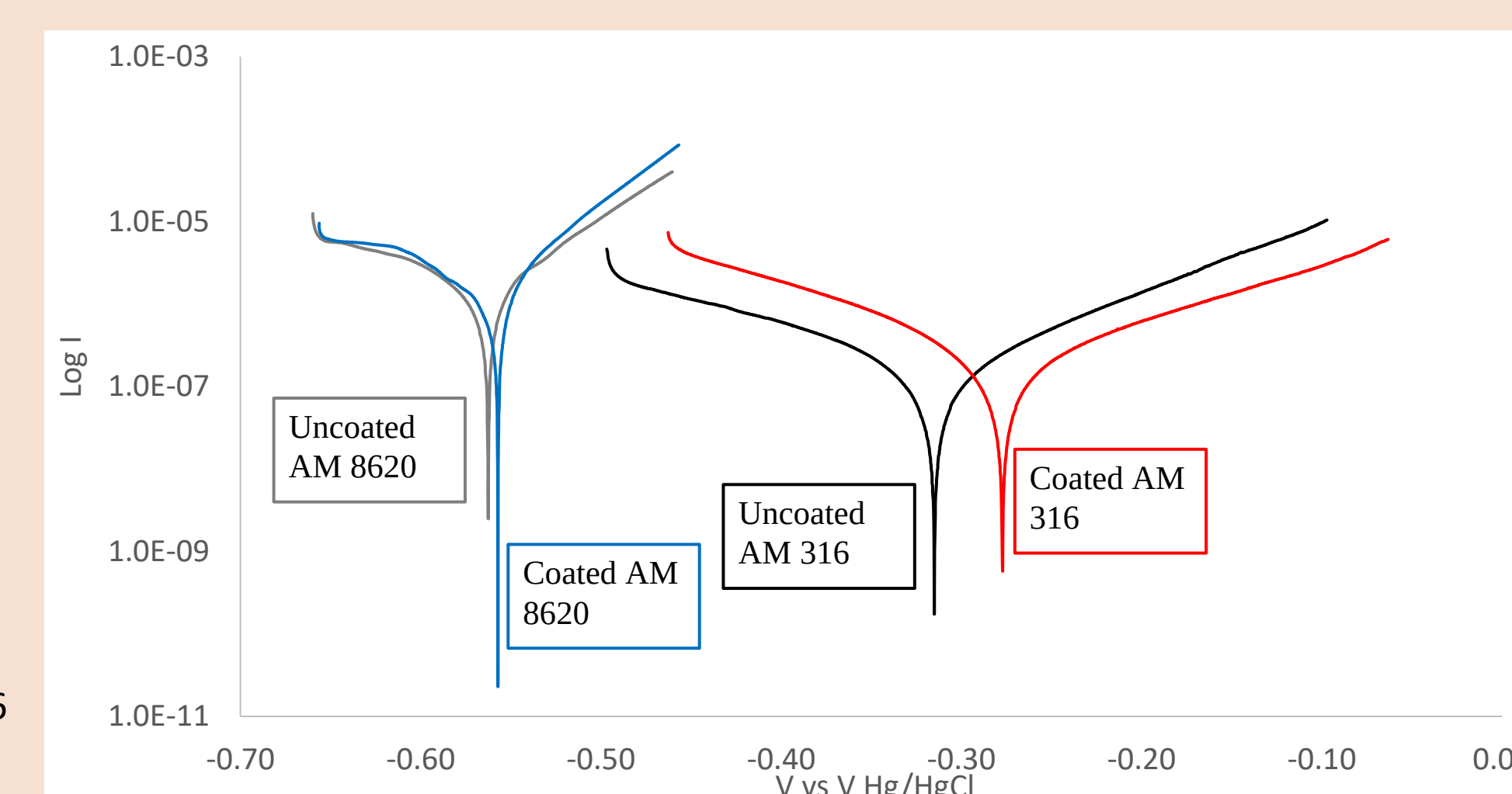


Polished AM8620 surface prepared for EGF or CVD graphene coating.

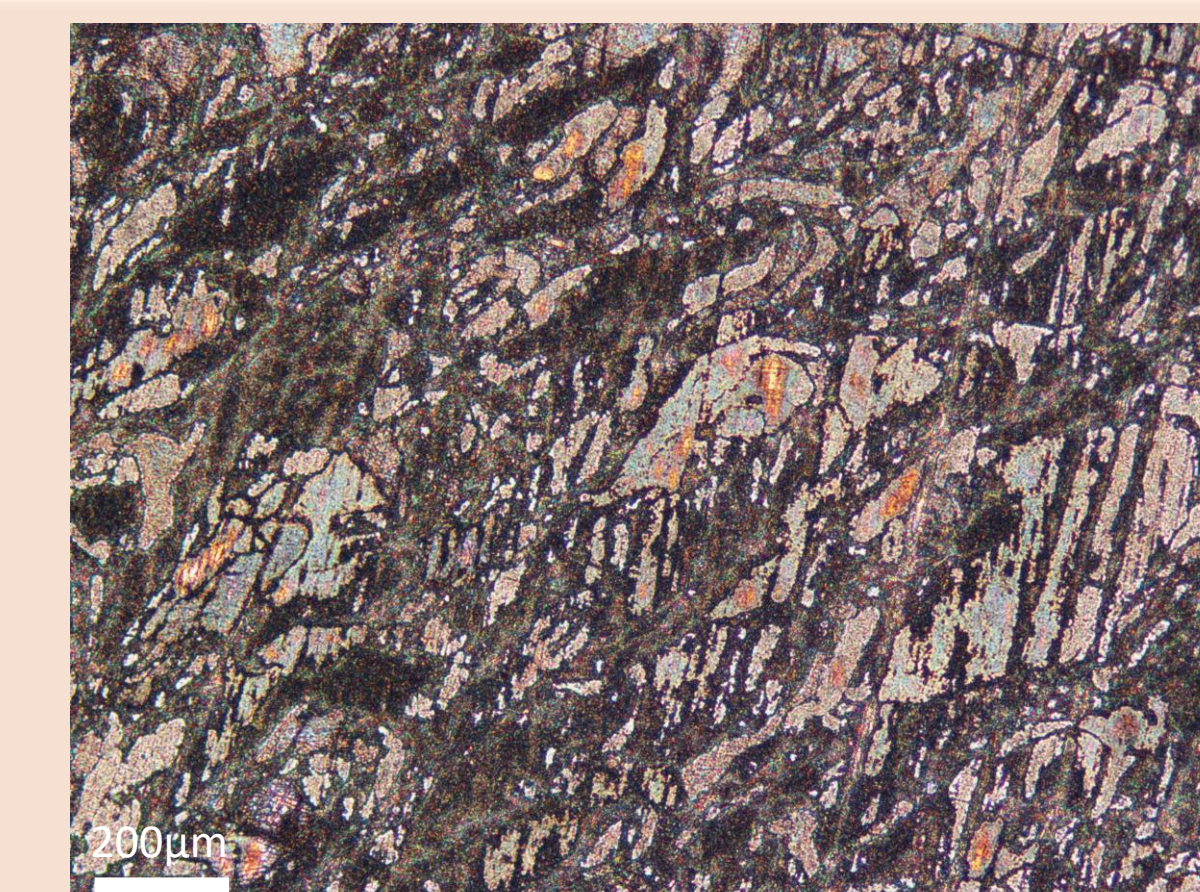
Conclusion & Future Work

Future work will focus on the development of graphene surface coatings for other additively manufactured and conventionally manufactured metals. Corrosion testing methods including Tafel plots and weight loss testing show promise for EGF coatings on 8620 and 316L steel, with future electrochemical testing to include CVD graphene coatings and EIS measurements.

A publication on the EFG coatings applied to AM8620, AM316, CM8620 and CM316 can be found in the journal *Results in Surfaces and Interfaces* by Kaleb Hood <https://doi.org/10.1016/j.rsufi.2023.100116>



Electrochemical testing (Tafel plots) on AM 316L and AM 8620 with and without EGF coatings.



Optical and SEM image of CVD graphene growth on additive manufactured 316L steel

